

Challenge: Speed Run

Points: 100 • Estimated Time: 60 Minutes

Challenge Scope

The robot must travel a straight 3-meter track and stop past the finish line in the shortest possible time.

Instructions & Engineering Hints

- **Objective:** Program and tune your robot to sprint across a straight 3-meter track and come to a complete stop past the finish line in the absolute shortest time possible.
- **Structural Considerations:** Focus on maximizing acceleration and traction. Consider the center of mass—if your robot is too top-heavy, it might tip over or wheel-spin when launching at full power. Secure your wheels tightly to prevent slip during rapid acceleration.
- **Sensor & Stopping Strategy:** Since speed is the priority, how will you know when to brake? Consider whether an optical color sensor looking at the floor or a time-based delay (running motors at max speed for a calculated duration) is more reliable for crossing the line without overshooting too far into the wall.
- **Iterative Testing:** Start by running your robot at 50% power to map its behavior, then gradually increase motor power. Measure your times across multiple trials to find the sweet spot between maximum speed and reliable stopping.